

SALEM, OR, USA

WMO: 726940

Lat: 44.905N

Lon: 123.001W

Elev: 205

StdP: 14.59

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24232

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	24.7	27.8	13.9	11.2	29.2	19.7	14.9	33.3	28.1	50.0	25.5	50.5	3.5	340	0.516

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.9	92.6	66.8	88.5	65.9	84.6	64.5	68.6	88.2	67.1	85.4	65.5	82.3	9.2	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.5	82.1	73.3	59.9	77.6	72.3	58.6	73.9	71.0	32.8	88.6	31.5	85.3	30.3	82.3	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.9	18.1	16.0	DB	18.5	100.0	3.9	3.6	15.7	102.6	13.4	104.7	11.2	106.7	8.4	109.3
			WB	17.4	71.8	4.1	1.8	14.4	73.1	12.1	74.1	9.8	75.1	6.8	76.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.9	41.9	43.5	47.1	50.8	57.1	62.3	69.0	68.9	63.7	54.0	46.3	41.1
	DBStd	11.32	6.20	5.42	4.91	5.22	5.87	5.74	5.36	4.99	5.36	5.37	6.06	6.51
	HDD50	1065	258	188	116	49	7	0	0	0	0	23	141	283
	HDD65	4436	716	601	553	426	254	118	18	17	88	344	561	740
	CDD50	2482	7	8	28	74	228	370	590	586	410	145	29	7
	CDD65	379	0	0	0	1	10	38	143	138	48	1	0	0
	CDH74	5081	0	0	0	39	246	610	1809	1635	708	34	0	0
	CDH80	2028	0	0	0	5	68	229	792	681	248	5	0	0
	WSAvg	6.3	6.7	6.8	7.2	6.6	6.0	6.2	6.2	5.7	5.3	5.6	6.7	7.0
	PrecAvg	39.3	6.0	4.5	4.2	2.6	2.0	1.4	0.5	0.7	1.4	2.9	6.3	6.6
(15) Precipitation	PrecMax	55.4	13.5	11.4	8.5	5.6	5.6	4.2	2.6	4.2	4.0	6.5	15.3	12.4
	PrecMin	23.7	0.2	0.7	0.9	0.6	0.0	0.3	0.0	0.0	0.0	0.1	1.1	1.3
	PrecStd	7.4	3.0	2.6	2.0	1.2	1.1	0.9	0.6	0.9	1.1	1.7	3.0	3.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.2	61.7	70.5	79.3	86.9	93.9	98.9	96.9	92.3	78.3	64.5	59.4
		MCWB	55.0	52.1	55.7	60.9	64.1	67.6	69.5	67.4	65.1	59.7	58.4	54.7
	2%	DB	55.9	57.9	64.6	72.2	80.9	86.4	92.9	91.8	86.6	72.6	60.5	55.7
		MCWB	51.6	50.5	52.8	57.6	62.3	65.1	67.6	66.4	63.6	57.7	55.7	52.5
	5%	DB	53.4	55.0	60.1	66.3	75.4	80.9	88.3	87.4	82.2	67.8	57.7	53.0
		MCWB	49.8	49.5	51.2	54.4	60.3	63.0	66.5	65.7	62.7	56.7	53.6	50.3
	10%	DB	51.2	52.5	56.6	61.6	70.3	75.9	84.0	83.2	77.4	63.7	55.3	50.2
		MCWB	48.2	47.9	49.9	52.2	57.7	60.9	64.9	64.4	61.4	55.7	51.7	47.5
	0.4%	WB	56.3	55.4	57.3	62.2	66.7	69.1	71.6	69.9	68.3	63.8	60.2	55.9
		MCDB	58.4	58.4	66.0	76.6	82.6	89.9	93.9	91.9	85.8	71.6	63.4	58.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	52.8	52.7	54.6	58.4	63.6	66.1	69.0	68.1	66.0	60.5	57.0	53.0
		MCDB	54.9	55.7	61.4	70.2	77.5	84.0	88.7	88.1	81.6	67.5	59.6	55.1
	5%	WB	50.4	50.4	52.5	55.4	61.0	63.9	67.3	66.5	64.1	58.5	54.4	50.6
		MCDB	52.8	54.0	58.5	64.1	73.4	79.2	86.0	84.6	78.2	64.4	56.7	52.8
	10%	WB	48.3	48.2	50.7	53.0	58.6	61.7	65.4	64.9	62.2	56.8	52.3	47.9
		MCDB	50.9	51.8	55.8	60.2	69.2	74.4	82.6	81.4	74.9	62.3	54.6	50.1
(35) Mean Daily Temperature Range	5% DB	MDBR	11.2	14.6	16.7	18.6	20.6	22.1	26.9	26.7	24.8	18.7	13.0	10.7
		MCDBR	13.1	18.7	24.4	29.4	31.8	32.5	34.5	33.9	34.1	27.1	14.7	11.5
		MCWBR	9.4	12.0	14.2	16.2	16.0	14.3	13.2	12.8	14.2	14.0	9.8	9.0
	5% WB	MCDBR	11.0	14.2	20.3	25.5	28.2	30.4	32.3	31.1	29.5	19.5	11.5	10.4
		MCWBR	8.8	10.1	12.7	14.8	14.9	14.0	13.3	12.5	13.2	11.4	8.5	8.8
(40) Clear-Sky Solar Irradiance	taub		0.316	0.316	0.326	0.355	0.365	0.358	0.350	0.353	0.343	0.330	0.324	0.322
	taud		2.535	2.542	2.518	2.400	2.391	2.418	2.462	2.472	2.492	2.535	2.517	2.489
	Ebn at Noon		256	279	290	288	287	288	289	285	279	268	249	239
	Edh at Noon		21	25	29	35	37	36	34	32	30	25	21	20
(44) All-Sky Solar Radiation	RadAvg		397	656	957	1384	1761	1963	2244	1920	1433	839	455	322
	RadStd		63	91	123	124	175	177	143	122	122	98	61	49

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+68
(47) Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+164	+40

Nomenclature: See separate page